POWER INPUT_1

PWR_FLAG
Q1
IRF4905
VOUT
R1
100k
R2
100k
C10
1000uF/63V
C65
1000uF/63V
C12
1000uF/63V
F1
Fuse
22.2V
LIPO_INPUT1
Conn_01x02_Socket
SW1
SW_SPDT
R_LED1
10k
D2
LED
GND

JETSON POWER_2

VOUT
C27
220uF/50V
C28
105/50V
C29
105/50V
JETSON_NANO_1
XL4015
VIN
SW
VC
GND
FB
FB_JN
L1
47uH
D7
B560
C30
105/25V
C33
330uF/25V
R12
10k
C34
33nF/25V
R13
3.3k
JETSON_JACK1
Barrel_Jack_Switch
GND

POWER 5V_3

VOUT
C43
220uF/50V
C45
105/50V
C49
105/50V
PERIPHERAL_BUCK(5V)1
XL4015
L2
47uH
D8
B560
C53
105/25V
C56
330uF/25V
R31
10k
C61
33nF/25V
R32
3.3k
FB_5
VOUT2_5V_5A
GND

POWER 12V_4

VOUT
C44
220uF/50V
C46
105/50V
C50
105/50V
PERIPHERAL_BUCK(12V)1
XL4015
L3
47uH
D9
B560
C54
105/25V
C57
330uF/25V
R33
13k
C62
33nF/25V
R34
1.5k
FB_12
VOUT4_12V_4A
GND

POWER FOR MCU(2)_20

VOUT3_3.3V_1A
PWR_FLAG
VBAT
C1
4.7uF/25V
D12
BAT54C
BT1
Battery_Cell
C63
104/25V
C4
104/25V
C9
104/25V
C11
104/25V
C14
104/25V
C17
104/25V
C19
104/25V
VCAP1
VCAP2
C23
2.2uF/25V
C26
2.2uF/25V
GND

SYS_WKUP_MODE_22

VOUT3_3.3V_1A
SW3
SW_Push
SYS_WKUP
R42
10k
GND

MCU STM32F407VET6

RAM: 196K
FLASH: 512K
ARM32_Cortex-M3 CPU
168 MHz

VBAT
VDD3_3.3V_1A
VDDA
NRST
BOOT0
VREF+
OSC_P
OSC_N
EXTI7_BAROMETER
PE8_OUTPUT_BAROMETER
PD13_TIM4_CH2
PD13_TIM4_CH1
VCAP1
VCAP2
SYS_WKUP
PA1_TIM2_CH2
PA2_TIM2_CH3
PA3_TIM2_CH4
PA5_TIM2_CH1
PA8_OUTPUT_TRANSCEIVER
USB_OTG_FS_VBUS
USB_OTG_FS_N
USB_OTG_FS_P
SWCLK
SWDIO
PB0_TIM3_CH3
PB1_TIM3_CH4
EXTI2_IMU
SPI1_SCLK_IMU
SPI1_MISO_IMU
SPI1_MOSI_IMU
PB6_OUTPUT_IMU
PB7_OUTPUT_IMU
SPI2_SCK_BAROMETER
CAN2_RX
CAN2_TX
SPI2_MISO_BAROMETER
SPI2_MOSI_BAROMETER
PC6_TIM3_CH1
PC7_TIM3_CH2
OSC32_P
OSC32_N
U1
STM32F407VETx
GND

POWER FOR MCU(1)_20

VOUT3_3.3V_1A
PWR_FLAG
FB1
FerriteBead
VDDA
C2
105/25V
C5
105/25V
R41
1k
D11
LED
GND

SYS_CLOCK_19

OSC_P
OSC_N
R40
1M
Y1
8M
C15
22pF/25V
C18
22pF/25V
GND

BOOT_18

VOUT3_3.3V_1A
R6
10k
NRST
BOOT0
SW2
SW_Push
R7
10k
C22
105/25V
GND

RTC_21

OSC32_P
OSC32_N
Y2
32.768K
C67
12pF/25V
C68
12pF/25V
GND

SWD_17

DEBUG_1
Conn_01x04_Pin
VOUT3_3.3V_1A
SWCLK
SWDIO
GND

USB PROTOCOL_15

J2
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D-
D-
D+
D+
SBU1
SBU2
SHIELD
GND
SH
A1
USB_OTG_FS_VBUS
R10
5.1k
R11
5.1k
R9
22
R8
22
USB_OTG_FS_N
USB_OTG_FS_P
FB2
FerriteBead
VOUT3_3.3V_1A
ESD_PROTECTION1
USBLC6-2SC6
USB_OTG_FS_P
USB_OTG_FS_N
GND

CAN_PROTOCOL 16

CAN_H
CAN_L
GND
CAN_1
Screw_Terminal_01x03
VOUT3_3.3V_1A
C3
106/25V
C8
105/25V
TRANSCEIVER1
SN65HVD230
CAN2_TX
CAN2_RX
PA8_OUTPUT_TRANSCEIVER
D
R
Vref
Rs
VCC
GND
CANH
CANL
R3
60
R4
60
C16
47nF/25V
D3
SM712_SOT23
CAN_H
CAN_L
GND

RC SERVO_22

PA3_TIM2_CH4
VOUT2_5V_5A
C73
106/25V
C76
104/25V
PWM
M1
Motor_Servo
PA2_TIM2_CH3
VOUT2_5V_5A
C79
106/25V
C82
104/25V
PWM
M2
Motor_Servo
GND

IMU SIGNAL_14

VOUT3_3.3V_1A
C64
106/25V
C6
104/25V
C13
104/25V
VCCA_LS_1.8V
C66
106/25V
R5
10K
C20
104/25V
VOUT3_3.3V_1A
VDDIO_IMU_1.8V
C24
104/25V
SPI1_MISO_IMU
SPI1_MOSI_IMU
SPI1_SCLK_IMU
PB6_OUTPUT_IMU
PB7_OUTPUT_IMU
EXTI2_IMU
VCCB
VCCA
OE
B1
B2
B3
B4
B5
B6
B7
B8
A1
A2
A3
A4
A5
A6
A7
A8
GND
LOGIC_SHIFTER1
TXS0108EPW
SDO/AD0
SDA/SDI
SCL/SCLK
CS
FSYNC
INT1
VDDIO
VDD
AUX_DA
AUX_CL
REGOUT
IMU_1
ICM-20948
C25
104/25V
GND

POWER IMU_13

D4
B360
VOUT3_3.3V_1A
ICM_LDO2
AMS1117-1.8
VI
VO
GND
C70
106/25V
C72
104/25V
C75
106/25V
C21
104/25V
VDDIO_IMU_1.8V
GND

POWER 3.3V_11

D6
B360
VOUT2_5V_5A
MCU_LDO_1
AMS1117-3.3
VI
VO
GND
C77
106/25V
C80
104/25V
C83
106/25V
C31
104/25V
VOUT3_3.3V_1A
GND

POWER LS A_12

D5
B360
VOUT3_3.3V_1A
ICM_LDO1
AMS1117-1.8
VI
VO
GND
C78
106/25V
C81
104/25V
C84
106/25V
C32
104/25V
VCCA_LS_1.8V
GND

DECOUPLING FOR DRIVERS_5

VOUT4_12V_4A
C37
104/25V
C40
47uF/25V
GND
DRIVER_LEFT1
DRV8870DDA
PA1_TIM2_CH2
PB0_TIM3_CH3
VOUT3_3.3V_1A
IN1
IN2
VM
OUT1
OUT2
VREF
GND
ISEN
LMotor_12V
LMotor−
R14
0.2
C41
104/25V
C42
47uF/25V
DRIVER_RIGHT1
DRV8870DDA
PA5_TIM2_CH1
PB1_TIM3_CH4
RMotor_12V
RMotor−
R15
0.2

CONTROL DRIVERS_6

DECOUPLING FOR BAROMETER_9

VOUT3_3.3V_1A
C35
106/25V
C38
105/25V
C36
106/25V
C39
105/25V
GND

BAROMETER_SIGNAL_10

VOUT3_3.3V_1A
VOUT3_3.3V_1A
BAROMETER_1
LPS22HB
VDD
Vdd_IO
SDA
SAO
SCL
CS
INT_DRDY
GND
SPI2_MOSI_BAROMETER
SPI2_MISO_BAROMETER
SPI2_SCK_BAROMETER
PE8_OUTPUT_BAROMETER
EXTI7_BAROMETER

ENCODER FEEDBACK_8

PD13_TIM4_CH1
PD13_TIM4_CH2
VOUT2_5V_5A
REncoder_A
REncoder_B
C85
106/25V
C87
105/25V
REncoder−
PC6_TIM3_CH1
PC7_TIM3_CH2
LEncoder_A
LEncoder_B
C86
106/25V
C88
105/25V
LEncoder−
GND

MOTOR_PINOUT_7

LMotor_12V
LMotor−
MOTOR_LEFT1
Screw_Terminal_01x02
VOUT2_5V_5A
LEncoder_A
LEncoder_B
LEncoder−
ENCODER_LEFT1
Screw_Terminal_01x04
RMotor_12V
RMotor−
MOTOR_RIGHT1
Screw_Terminal_01x02
VOUT2_5V_5A
REncoder_A
REncoder_B
REncoder−
ENCODER_RIGHT1
Screw_Terminal_01x04

IMU pins I/Os operate normal at VDDIO = 1.8V, T = 25°C
Input Power: 22.2V LiPo (6S). Max voltage: 25.2V
No copper zone for IMU when layout PCB
Sensor fusion with IMU and barometer

UAVL
Sheet: /
File: Mobile_robot.kicad_sch

Title: Mobile robot

Size: A2	Date: 2026-03-27	Rev: Up to date
KiCad E.D.A. 10.0.0		Id: 1/1